Semester	B.E. Semester VII – Computer Engineering
Subject	DevOps Lab
Subject In-charge	Dr. Snehal Andhare

Student Name	Vicky Pukale
Roll Number	23102A0033
Grade and Subject Teacher's Signature	

Task Number	5
Project Title:	Secure CI/CD Pipeline for a Loan Application Tracking Portal
Task Title	Feature Development with Branching: Implement the first core workflow of the Loan Application Tracking Portal on a feature branch. Use add, commit, push, pull and logs. Raise a pull request, review changes and merge into the development branch.
Deliverable:	**All code files for Feature 1: Loan Application Intake & Event Entry have been created in project workspace:**

Created Feature 1 Files

1. `LoanApplication.java` - JPA Entity with fields (`applicationId` , `applicantName` , `loanAmount` , `loanType` , `currentStatus` , etc.).
2. `LoanEvent.java` - Event logging entity for chronological stage tracking.
3. `LoanApplicationRepository.java` - Data access repository with search and count queries.
4. `LoanEventRepository.java` - Repository for event history.
5. `LoanService.java` - Service layer auto-generating `LOAN-1001` format IDs and logging initial `SUBMITTED` events.
6. `LoanController.java` - MVC endpoints for `/apply` form submission and `/dashboard` view.
7. `apply.html` - Responsive intake form with explicit HTML IDs for future Selenium automation.
8. `dashboard.html` - Initial dashboard table displaying submitted loan applications.
9. `LoanServiceTest.java` - Unit & integration test verifying ID generation and event persistence.

1. Create and Switch to Feature Branch:

```
D:\Vicky_Project\DevOps\DevOps_Project>git checkout develop
Already on 'develop'
Your branch is up to date with 'origin/develop'.

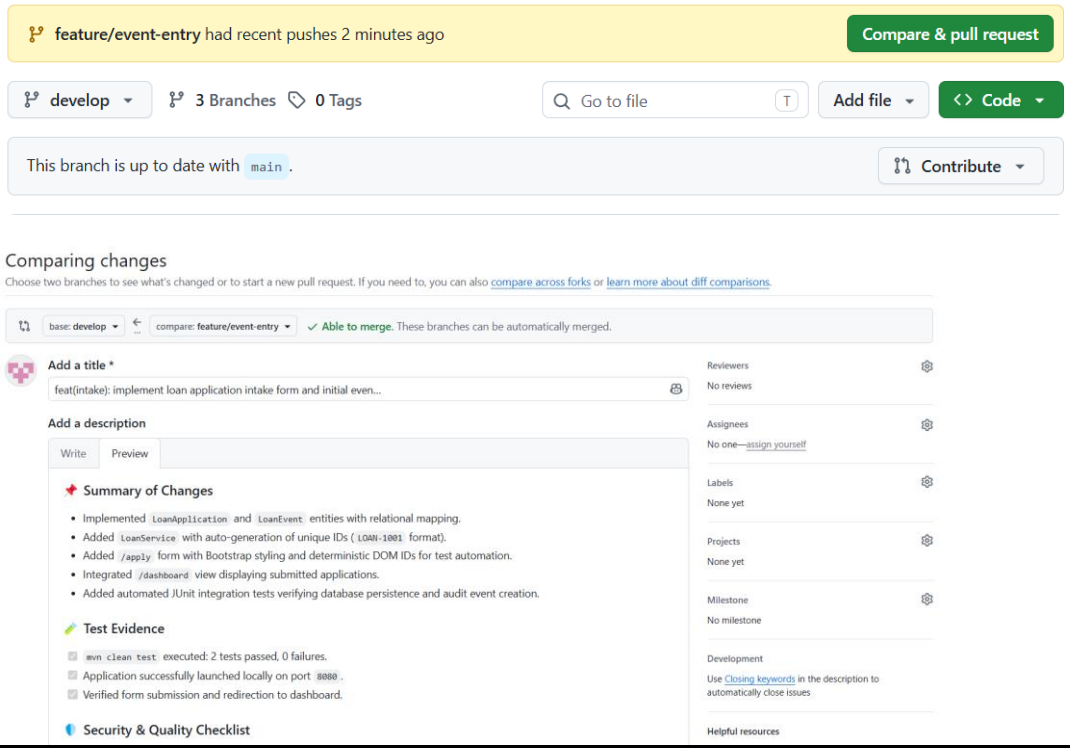
D:\Vicky_Project\DevOps\DevOps_Project>git checkout -b feature/event-entry
Switched to a new branch 'feature/event-entry'
```

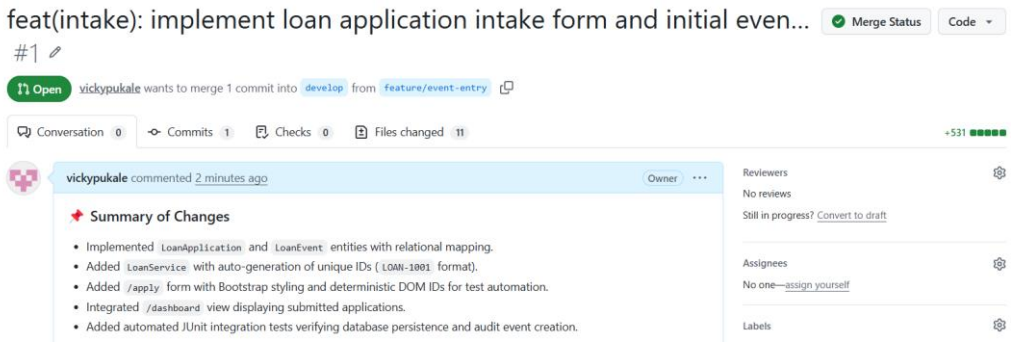
2. Stage, Commit, and Push Feature Branch:

```
D:\Vicky_Project\DevOps\DevOps_Project>git add .
warning: in the working copy of 'src/main/java/com/loanportal/controller/LoanController.java', LF will be replaced by CR
LF the next time Git touches it
warning: in the working copy of 'src/main/java/com/loanportal/model/LoanApplication.java', LF will be replaced by CRLF t
he next time Git touches it
warning: in the working copy of 'src/main/java/com/loanportal/model/LoanEvent.java', LF will be replaced by CRLF the nex
t time Git touches it
warning: in the working copy of 'src/main/java/com/loanportal/repository/LoanApplicationRepository.java', LF will be rep
laced by CRLF the next time Git touches it
warning: in the working copy of 'src/main/java/com/loanportal/repository/LoanEventRepository.java', LF will be replaced
by CRLF the next time Git touches it
warning: in the working copy of 'src/main/java/com/loanportal/service/LoanService.java', LF will be replaced by CRLF the
 next time Git touches it
warning: in the working copy of 'src/main/resources/templates/apply.html', LF will be replaced by CRLF the next time Git
 touches it
warning: in the working copy of 'src/main/resources/templates/dashboard.html', LF will be replaced by CRLF the next time
 Git touches it
warning: in the working copy of 'src/test/java/com/loanportal/service/LoanServiceTest.java', LF will be replaced by CRLF
 the next time Git touches it
D:\Vicky_Project\DevOps\DevOps_Project>git commit -m "feat(intake): implement loan application intake form and initial e
vent entry"
[feature/event-entry 3dd587b] feat(intake): implement loan application intake form and initial event entry
 11 files changed, 531 insertions(+)
 create mode 100644 DOC/DevOps_Task4_Vicky_Pukale_23102A0033.docx
 create mode 100644 PDF/DevOps_Task4_Vicky_Pukale_23102A0033.pdf
 create mode 100644 src/main/java/com/loanportal/controller/LoanController.java
 create mode 100644 src/main/java/com/loanportal/model/LoanApplication.java
 create mode 100644 src/main/java/com/loanportal/model/LoanEvent.java
 create mode 100644 src/main/java/com/loanportal/repository/LoanApplicationRepository.java
 create mode 100644 src/main/java/com/loanportal/repository/LoanEventRepository.java
 create mode 100644 src/main/java/com/loanportal/service/LoanService.java
 create mode 100644 src/main/resources/templates/apply.html
 create mode 100644 src/main/resources/templates/dashboard.html
 create mode 100644 src/test/java/com/loanportal/service/LoanServiceTest.java

D:\Vicky_Project\DevOps\DevOps_Project>git push -u origin feature/event-entry
Enumerating objects: 43, done.
Counting objects: 100% (43/43), done.
Delta compression using up to 16 threads
Compressing objects: 100% (23/23), done.
Writing objects: 100% (31/31), 1.03 MiB | 279.00 KiB/s, done.
Total 31 (delta 4), reused 0 (delta 0), pack-reused 0 (from 0)
remote: Resolving deltas: 100% (4/4), completed with 3 local objects.
remote:
remote: Create a pull request for 'feature/event-entry' on GitHub by visiting:
remote:      https://github.com/vickypukale/loan-application-tracking-portal/pull/new/feature/event-entry
remote:
To https://github.com/vickypukale/loan-application-tracking-portal.git
 * [new branch]      feature/event-entry -> feature/event-entry
branch 'feature/event-entry' set up to track 'origin/feature/event-entry'.
```

3. Raise a Pull Request (PR) on GitHub:





4. Update Local Repository & Verify Git Log:

```
D:\Vicky_Project\DevOps\DevOps_Project>git checkout develop
Switched to branch 'develop'
Your branch is up to date with 'origin/develop'.

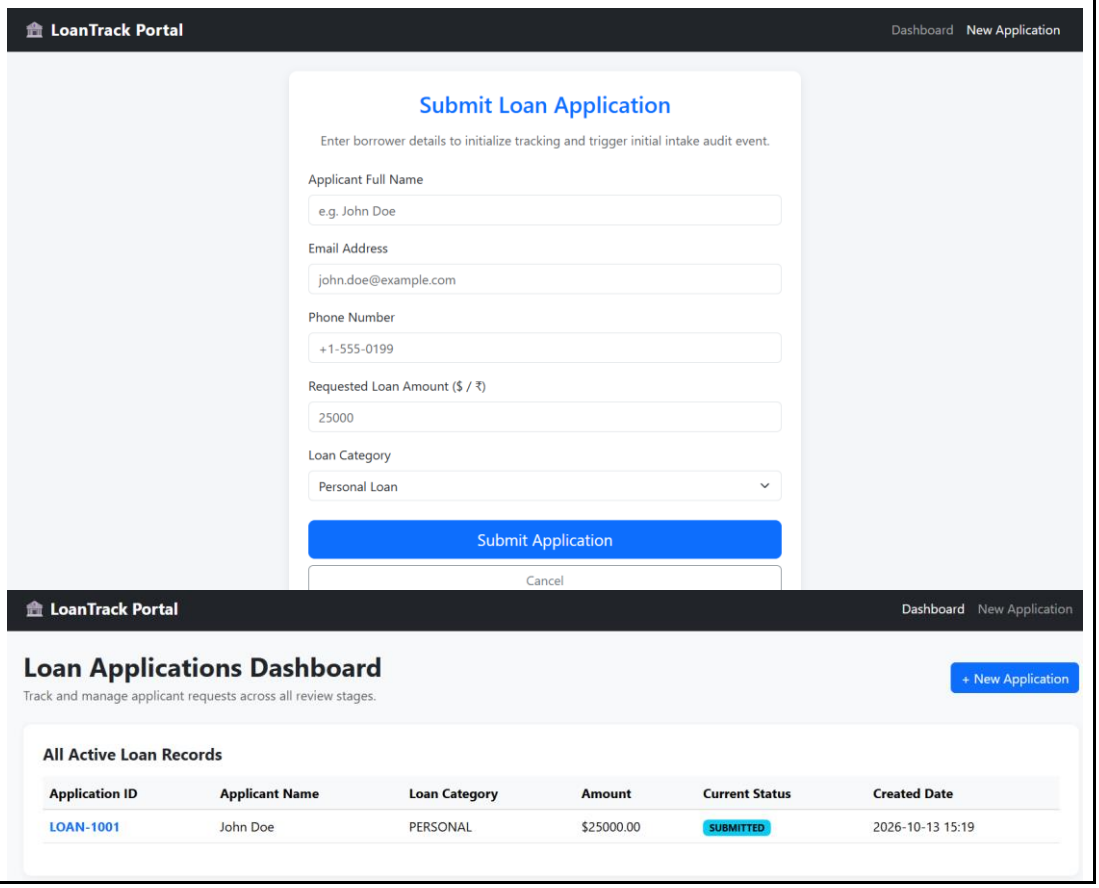
D:\Vicky_Project\DevOps\DevOps_Project>git pull origin develop
remote: Enumerating objects: 1, done.
remote: Counting objects: 100% (1/1), done.
remote: Total 1 (delta 0), reused 0 (delta 0), pack-reused 0 (from 0)
Unpacking objects: 100% (1/1), 953 bytes | 953.00 KiB/s, done.
From https://github.com/vickypukale/loan-application-tracking-portal
 * branch            develop    -> FETCH_HEAD
   d9c66ab..4b501a6  develop    -> origin/develop
Updating d9c66ab..4b501a6
Fast-forward
 DOC/DevOps_Task4_Vicky_Pukale_23102A0033.docx      | Bin 0 -> 490920 bytes
 PDF/DevOps_Task4_Vicky_Pukale_23102A0033.pdf       | Bin 0 -> 631979 bytes
 .../com/loanportal/controller/LoanController.java  |  44 +++++++++
 .../java/com/loanportal/model/LoanApplication.java | 100 ++++++++++++++++++++++
 src/main/java/com/loanportal/model/LoanEvent.java  |  51 ++++++++++
 .../repository/LoanApplicationRepository.java      |  16 ++++
 .../loanportal/repository/LoanEventRepository.java |  13 +++
 .../java/com/loanportal/service/LoanService.java   |  99 ++++++++++++++++++++
 src/main/resources/templates/apply.html            |  74 ++++++++++++++++
 src/main/resources/templates/dashboard.html        |  81 ++++++++++++++++
 .../com/loanportal/service/LoanServiceTest.java    |  53 ++++++++++
 11 files changed, 531 insertions(+)
 create mode 100644 DOC/DevOps_Task4_Vicky_Pukale_23102A0033.docx
 create mode 100644 PDF/DevOps_Task4_Vicky_Pukale_23102A0033.pdf
 create mode 100644 src/main/java/com/loanportal/controller/LoanController.java
 create mode 100644 src/main/java/com/loanportal/model/LoanApplication.java
 create mode 100644 src/main/java/com/loanportal/model/LoanEvent.java
 create mode 100644 src/main/java/com/loanportal/repository/LoanApplicationRepository.java
 create mode 100644 src/main/java/com/loanportal/repository/LoanEventRepository.java
 create mode 100644 src/main/java/com/loanportal/service/LoanService.java
 create mode 100644 src/main/resources/templates/apply.html
 create mode 100644 src/main/resources/templates/dashboard.html
 create mode 100644 src/test/java/com/loanportal/service/LoanServiceTest.java

D:\Vicky_Project\DevOps\DevOps_Project>git log --oneline --graph --decorate -n 10
*   4b501a6 (HEAD -> develop, origin/develop) Merge pull request #1 from vickypukale/feature/event-entry
|\
| * 3dd587b (origin/feature/event-entry, feature/event-entry) feat(intake): implement loan application intake form and initial event entry
|/
* d9c66ab (origin/main, main) chore: initialize project skeleton with Maven, Spring Boot, and GitHub templates
```

5. Test the Working Feature Locally:

mvn spring-boot:run

```
D:\Vicky_Project\DevOps\DevOps_Project>mvn spring-boot:run
[INFO] Scanning for projects...
[INFO]
[INFO] ----------< com.loanportal:loan-application-tracking-portal >-----------
[INFO] Building Loan Application Tracking Portal 1.0.0-SNAPSHOT
[INFO]   from pom.xml
[INFO] --------------------------------[ jar ]---------------------------------
[INFO]
```


LoanTrack Portal
Dashboard
New Application
Submit Loan Application
Enter borrower details to initialize tracking and trigger initial intake audit event.
Applicant Full Name
e.g. John Doe
Email Address
john.doe@example.com
Phone Number
+1-555-0199
Requested Loan Amount ($ / ₹)
25000
Loan Category
Personal Loan
Submit Application
Cancel

LoanTrack Portal
Dashboard
New Application
Loan Applications Dashboard
Track and manage applicant requests across all review stages.
+ New Application
All Active Loan Records
Application ID
Applicant Name
Loan Category
Amount
Current Status
Created Date
LOAN-1001
John Doe
PERSONAL
$25000.00
SUBMITTED
2026-10-13 15:19